

Referee report: the o_ν socle case, offset coboundary, lock, and the $L \subsetneq \text{Aut}(D)$ torsor criterion

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Reviewed: your 09-30 o_ν *fixed- σ slices* (3pp, WIP fc3d742);
your 10-01 o_ν *socle / $L \subsetneq \text{Aut}$ tabulation* (4pp, WIP 648fa09);
emails UID 304, 305, 307, 311
WIP commit: PENDING (grandpa-rick/work-in-progress)

Source caveat. The 10-01 PDF is a self-described *tabulation* of [S2] (2026-09-30-general-non-trivial-onu.tex) and [S3] (2026-09-30-general-H-onu-offset.tex), which are local-only on your side (the GH_TOKEN issue). I have **not** seen S2 or S3. Everything below is re-derived from the definitions stated in the 10-01 PDF, not from your proofs. Where I say “gap” it may well be closed in S2/S3; please push them when you can.

Reproducibility. All computations were done with code written from scratch, now in the directory

grandpa-rick/work-in-progress/reviews/2026-10-02-macbeth-scripts/: braces.py (all skew braces with abelian additive group A , as regular subgroups of $\text{Hol}(A)$, by backtracking closure search), lock.tab.py $\rightarrow$ lock.tab.log, lock_counts.py $\rightarrow$ lock_counts.log, and sanity checks (sanity.log): 6 braces on $\mathbb{Z}/8$ (matching an independent brute force over all $\lambda : \mathbb{Z}/8 \rightarrow \text{Aut}(\mathbb{Z}/8)$), 28 on $\mathbb{Z}/2 \times \mathbb{Z}/4$, 16 on $\mathbb{Z}/16$, 160 on $\mathbb{Z}/2 \times \mathbb{Z}/8$; all pass the brace-axiom check.

Conventions and ambiguities

Skew brace $(G, +, \circ)$ with $a \circ (b + c) = a \circ b - a + a \circ c$, $\lambda_a(b) = -a + a \circ b$, $\lambda : (G, \circ) \rightarrow \text{Aut}(G, +)$. In all witnesses $(G, +)$ is abelian, so $\mu \equiv \text{id}$; I work only in that setting. D is an ideal, $H = G/D$, s a normalised set-section ($s(0) = 0$). Additive 2-cocycle $\beta(k, k') := s(k) + s(k') - s(k + k') \in D$. For $H = \mathbb{Z}/2$: $u := s(1)$, $b := \beta(1, 1) = 2u$.

$$\nu_h := \lambda_{s(h)}|_D, \quad \hat{\sigma}_h(x) := s(h)^{-1} \circ x \circ s(h) \text{ (}\circ\text{-inverse)}, \quad \delta_h(x) := \nu_h \hat{\sigma}_h(x) - x.$$

Check: $\nu_h \hat{\sigma}_h(x) = -s + s \circ s^{-1} \circ x \circ s = -s + x \circ s$ and $\lambda_x(s) - s = -x + x \circ s - s$, so $\delta_h(x) = \lambda_x(s(h)) - s(h)$ (uses $(G, +)$ abelian). This matches your “[S2, Lem. 3.1]”.

A1 (“ $\nu \hat{\sigma} \bmod 4$ ”).

The PDF treats $\nu \hat{\sigma}$ as a unit of $\mathbb{Z}/8$. A priori $\nu \hat{\sigma} : x \mapsto x + \lambda_x(u) - u$ is only a permutation of D ; $x \mapsto \lambda_x(u)$ need not be additive in x . I adopt $c_0 := \nu \hat{\sigma}(g)$ for a fixed generator g of D (“ $\nu \hat{\sigma}$ at a generator”, which is how the PDF evaluates it), and separately test whether $\nu \hat{\sigma}$ is additive (Item 2).

A2 (σ_+).

The “additive σ_+ used in [S1]” is not defined in either PDF (with $(G, +)$ abelian, additive conjugation is trivial). I could not reproduce the 104/104 vs 80/104 comparison and do not grade it.

A3 (what is “ $\text{im } \varphi^+$ ” for a fixed triplet).

ν and $\hat{\sigma}$ depend on the section: replacing $s(h) \mapsto s(h) + t(h)$, $t(h) \in D$, replaces ν_h by $\nu_h \circ$ (element of L) and $\delta_h(d)$ by $\delta_h(d) + (\lambda_d - 1)t(h)$. So the triplet is defined only up to this action; I read “ $\text{im } \varphi^+$ for (ν, σ) ” as the set of classes $[\beta] \in H_+^2$ realised by braces whose section-class of triplet is the given one. For $H = \mathbb{Z}/2$, $D = \mathbb{Z}/8$, $L = \{1, 5\}$, the section invariant is $\nu\hat{\sigma} \bmod 4$ (verified, Item 2).

A4 (“socle”).

“Socle” is used both for “the coset L ” and for “ $\{1, 5\}$, the squares”. For $L = \{1, 5\}$ these agree; otherwise not. I test both readings in Item 3.

1 Offset coboundary $c^d = d_+[\delta_\bullet(d)]$ for general H

Claim (10-01 PDF, §§3–4, citing [S3, Lem. 2.1, Cor. 2.3]):

$$(\lambda_d^D - \text{id}) \beta(k, k') = \delta_k(d) + \delta_{k'}(d) - \delta_{k+k'}(d) \quad (d \in D, k, k' \in H), \quad (\star)$$

i.e. $c^d := (\lambda_d^D - 1)\beta = d_+[k \mapsto \delta_k(d)]$. Graded “proved, general H ”.

Independent re-derivation.

1. $s(k) + s(k') = \beta(k, k') + s(k + k')$ in $(G, +)$ (definition of β ; abelian).
2. Apply λ_d (additive on G): $\lambda_d s(k) + \lambda_d s(k') = \lambda_d \beta(k, k') + \lambda_d s(k + k')$.
3. Subtract 1: $[\lambda_d s(k) - s(k)] + [\lambda_d s(k') - s(k')] = (\lambda_d - 1)\beta(k, k') + [\lambda_d s(k + k') - s(k + k')]$.
4. With $\delta_k(d) = \lambda_d(s(k)) - s(k)$ this is exactly $(\star)$. Since $\beta \in D$ and D is λ -invariant, $\lambda_d \beta = \lambda_d^D \beta$. And $\delta_k(d) \in D$: $\lambda_d(g) - g = -d + g \circ (g^{-1} \circ d \circ g) - g = -d + \lambda_g(d') \in D$.
5. With μ trivial, $(d_+ f)(k, k') = f(k) + f(k') - f(k + k')$, so the RHS is $d_+[\delta_\bullet(d)](k, k')$.

Verdict. $(\star) / c^d = d_+[\delta_\bullet(d)]$ is **re-derived**; I upgrade it to **checked-sober**. It is really a four-line identity: additivity of λ_d applied to the cocycle relation. Two scope remarks:

- *Scope.* It needs $(G, +)$ abelian ($\mu \equiv \text{id}$). For non-abelian $(G, +)$ the d_+ must carry μ and the “subtract” step needs reordering. “General H ” in the PDF means general H with abelian $(G, +)$; I record it that way.
- *The offset is cochain-level and section-dependent.* Under $s \mapsto s + t$, $\delta_k(d) \mapsto \delta_k(d) + (\lambda_d - 1)t(k)$, hence $c^d \mapsto c^d + (\lambda_d - 1)d_+ t$. Its cohomology class is *always* 0 (it is a coboundary by $(\star)$), consistent with your Remark 4 that the cohomological condition $(\lambda_d - 1)_*[\beta] = 0$ is strictly weaker. So “ $c \neq 0$ ” must always be read as “ $c \neq 0$ as a cochain modulo the section action”, never as a class in H^2 .

Definition 2 is ill-typed as written. “ $o_\nu([\beta]) = (\lambda_1^D - 1)b - 2(\nu\hat{\sigma} - 1)$ on $H_+^2 = D/2D$ ”: $(\lambda_1 - 1)b$ is not a function of $b \bmod 2D$ in general. (It is for $D = \mathbb{Z}/8$, $\lambda_1 = 5$, since $4b \bmod 8$ only sees $b \bmod 2$.)

Gap: the general- H basepoint criterion attributed to [S3]. PDF Prop. 3: “ $0 \in \text{im } \varphi^+$ iff for every $d \in D$, $\delta_\bullet(d) : (H, +) \rightarrow (D, +)$ is a homomorphism ($c^d = 0$)”, graded proved for general H .

- $(\Rightarrow)$ If some brace realising the triplet has $[\beta] = 0$, choose the section with $\beta = 0$; then $(\star)$ gives $d_+\delta_\bullet(d) = 0$, so $\delta_\bullet(d)$ is a homomorphism *for that section*. Correct, modulo A3: the criterion must be phrased “for some section in the triplet class”.
- $(\Leftarrow)$ Does **not** follow from $(\star)$: $\delta_\bullet(d)$ a homomorphism for all d only gives $(\lambda_d - 1)\beta = 0$ for all d , i.e. β takes values in $D^L = \text{Fix}(L)$, not $\beta = 0$ or $[\beta] = 0$. Realising a split brace with the same triplet needs an existence argument that I cannot see in the tabulation. In the trivial-kernel case ($L = \{1\}$) the hypothesis is vacuous and $[\beta] \neq 0$ occurs (e.g. the trivial brace $\mathbb{Z}/16 \supset 2\mathbb{Z}/16$), so $(\Leftarrow)$ can only be true via the *existence of another brace* (here the trivial brace $\mathbb{Z}/2 \times \mathbb{Z}/8$) with the same triplet: it is a realisability theorem, not a corollary of $(\star)$. I keep this criterion at **peer-claimed** until I can read S3’s proof of $(\Leftarrow)$.

2 Lock $4b \equiv 2(\nu\hat{\sigma} - 1) \bmod 8$, $H = \mathbb{Z}/2$

Claim (10-01 PDF Thm. 1, citing [S2, Thm. 3.8]): for all $x \in D$, $(\lambda_x^D - 1)b = 2\delta_1(x)$, $b = 2u$; at a generator with $\lambda_1 = 5$ on $D = \mathbb{Z}/8$: $4b \equiv 2(\nu\hat{\sigma} - 1) \bmod 8$, and $[\beta] \neq 0 \iff \nu\hat{\sigma} \equiv 3 \bmod 4$. “Proved, 112/112.”

Proof check. Your derivation (axiom $x \circ (u + u) = x \circ u - x + x \circ u$) reduces, after writing $x \circ y = x + \lambda_x(y)$, to $\lambda_x(2u) = 2\lambda_x(u)$, i.e. additivity of λ_x . Then $(\lambda_x - 1)(2u) = 2(\lambda_x(u) - u) = 2\delta_1(x)$. So the lock in identity form is correct for every ideal D , $H = \mathbb{Z}/2$, $(G, +)$ abelian, and it is a one-line identity. Specialisation: on $D = \mathbb{Z}/8$ with $\lambda_g = \times 5$, $(5 - 1)b = 4b$ and $2\delta_1(g) = 2(c_0 - 1)$, $c_0 := \nu\hat{\sigma}(g)$; since $4b \bmod 8$ sees only $b \bmod 2$, b odd $\iff c_0 \equiv 3 \bmod 4$ (under reading A1).

Definitional gap (A1). The PDF treats $\nu\hat{\sigma}$ as an element of $\text{Aut}(D, +)$. But $\hat{\sigma}$ is $\circ$ -conjugation, an automorphism of $(D, \circ)$, not of $(D, +)$. Computed: in the $L = \{1, 5\}$ family $\hat{\sigma}$ is additive for every section (64/64 on $\mathbb{Z}/16$, 256/256 on $\mathbb{Z}/2 \times \mathbb{Z}/8$), so your reading is harmless there. For $L \in \{\{1, 3\}, \{1, 7\}, \{1, 3, 5, 7\}\}$, however, $\hat{\sigma}$ (and $\nu\hat{\sigma}$) is **non-additive for exactly half the sections** (192/384, 192/384, 256/512). So “ $\nu\hat{\sigma} \bmod 4$ ” and “ $\delta_h = \nu\hat{\sigma} - \text{id}$ is the defect *automorphism*” make sense only on $L = \{1, 5\}$; elsewhere δ_1 must be kept as a map $D \rightarrow D$.

Independent computation. For every brace, every cyclic ideal D of index 2, every section $u \notin D$, `lock.tab.py` checks the lock identity for all $x \in D$, the lock at a generator, additivity of ν , $\hat{\sigma}$, $\nu\hat{\sigma}$, and the parity of b by coset of $\nu\hat{\sigma}$ in $\text{Aut}(D)/L$. `lock.counts.py` reconciles your counts.

- **Lock identity** $(\lambda_x - 1)b = 2\delta_1(x)$: **100% of (brace, D , u) triples** in all four groups ($\mathbb{Z}/16$: 128/128; $\mathbb{Z}/2 \times \mathbb{Z}/8$: 1792/1792; $\mathbb{Z}/2 \times \mathbb{Z}/4$: 128/128; $\mathbb{Z}/8$: 24/24).
- **Count reconciliation, reproduced exactly.** Your units are (brace, D) pairs with D a *fixed* cyclic index-2 subgroup and L non-trivial: $\mathbb{Z}/16$: 8 (all $L = \{1, 5\}$); $\mathbb{Z}/2 \times \mathbb{Z}/8$: 96 per D ($= 24$

with $L = \{1, 3\}$, 32 with $L = \text{Aut}$, 16 with $L = \{1, 5\}$, 24 with $L = \{1, 7\}$; $\mathbb{Z}/2 \times \mathbb{Z}/4$: 8 per D ($L = \{1, 3\} = \text{Aut}(\mathbb{Z}/4)$). Total $8 + 96 + 8 = \mathbf{112}$. The “ $|G| = 16$ relation 24/24 (8 on $\mathbb{Z}/16$, 16 on $\mathbb{Z}/2 \times \mathbb{Z}/8$)” is the $L = \{1, 5\}$ pairs: $8 + 16 = \mathbf{24}$. $[\beta] \neq 0$ on all 8 $\mathbb{Z}/16$ pairs and $[\beta] = 0$ on all 96 $\mathbb{Z}/2 \times \mathbb{Z}/8$ pairs.

- **09-30 Table 2 reproduced exactly** with $\sigma := \hat{\sigma}$ ($L = \{1, 5\}$): the realised $(\hat{\sigma}, \nu)$ pairs are exactly your 12 rows ($\hat{\sigma} = 3, 7$ occur only with $\nu \in \{3, 7\}$), and b is odd exactly on the four rows with $\nu\hat{\sigma} \equiv 3 \pmod{4}$ (16 triples each), even on the other eight (32 each).
- Section invariance: for $L = \{1, 5\}$, $\nu\hat{\sigma} \pmod{L} = \nu\hat{\sigma} \pmod{4}$ is constant over the sections of a brace.
- Not reproduced: $\hat{\sigma}$ vs σ_+ (104/104 vs 80/104; σ_+ undefined, A2); “33/33”, “16/16”, “2/16 $d^2 \neq 0$ ” (operator-level objects not defined in the PDFs).

Verdict. Re-derived; I upgrade o-nu-lock-4b-mod8-H-Z2 to **checked-sober**, with scope “ $(G, +)$ abelian, $D = \mathbb{Z}/8$, $L = \{1, 5\}$, $\nu\hat{\sigma}$ read as a unit (it is additive in this family)”. The identity form is checked-sober for all ideals at $H = \mathbb{Z}/2$. The lock is a one-line consequence of additivity of λ_x , and $\nu\hat{\sigma} \notin \text{Aut}(D, +)$ in general (half of the sections for $L \ni 3$).

3 The $L \subsetneq \text{Aut}(D)$ torsor criterion, both directions

Claim (10-01 PDF Prop. 3, §5 table, summary): $\text{im } \varphi^+$ is a torsor without basepoint $\iff c_{[\nu], \sigma} = 2(\nu\hat{\sigma} - 1) \neq 0 \iff L \subsetneq \text{Aut}(D)$ and $\nu\hat{\sigma}$ non-socle; “torsor-without-basepoint requires $L \subsetneq \text{Aut}(D)$; when $L = \text{Aut}(D)$ the obstruction is linear and 0 is always realisable.”

Necessity: no basepoint $\Rightarrow L \subsetneq \text{Aut}(D)$

Your evidence is the $\mathbb{Z}/4$ hand computation (units of $\mathbb{Z}/4$ are odd, so $2(\nu\hat{\sigma} - 1) \equiv 0 \pmod{4}$), valid for $D = \mathbb{Z}/4$ only. No general argument is given, so the status in the PDF is: claimed in general, hand-checked for $D = \mathbb{Z}/4$. My computation: $D = \mathbb{Z}/8$ with $L = \text{Aut}(\mathbb{Z}/8)$ exists (32 pairs per D on $\mathbb{Z}/2 \times \mathbb{Z}/8$; none on $\mathbb{Z}/16$), and b is even on all 512 triples, so every triplet class is based. Also $D = \mathbb{Z}/4$, $L = \text{Aut}$: b even 64/64. So necessity is **computed** for $D \in \{\mathbb{Z}/4, \mathbb{Z}/8\}$, $H = \mathbb{Z}/2$. It is not proved in general.

Sufficiency: $L \subsetneq \text{Aut}(D)$ and $\nu\hat{\sigma}$ non-socle (or $c \neq 0$) $\Rightarrow$ no basepoint

This is false as stated. Counterexample (computed by our enumeration). Take $D = \mathbb{Z}/8$ with $L = \{1, 3\}$ (24 pairs per D on $\mathbb{Z}/2 \times \mathbb{Z}/8$; no such brace on $\mathbb{Z}/16$). The lock at the generator reads $2b \equiv 2(c_0 - 1) \pmod{8}$, so $b \equiv c_0 - 1 \pmod{4}$ is always even; computed, b is even on **all 384 triples**. Yet $L \subsetneq \text{Aut}(D)$, and

- reading “socle = L ”: the coset $\nu\hat{\sigma} \cdot L = \{5, 7\}$ occurs (96 triples, additive sections), is non-socle, and is still based;
- reading “socle = $\{1, 5\}$ ”: $\nu\hat{\sigma} = 3$ or 7 occurs, offset $c = 2(c_0 - 1) = 4 \neq 0$, still based.

The same holds for $L = \{1, 7\}$: cosets $\{1, 7\}$, $\{3, 5\}$, offsets $\{0, 4\}$, b even 384/384. So the counterexample stands under *both* readings of “socle”.

Also: $L = \{1\}$ (trivial-kernel brace) with $\nu\hat{\sigma} = 5 \notin L$ occurs on both $\mathbb{Z}/16$ (b odd) and $\mathbb{Z}/2 \times \mathbb{Z}/8$ (b even), so that triplet class is based although it is “non-socle” under the reading “socle = L ”.

True on the $L = \{1, 5\}$ family. There “socle = $L = \{1, 5\}$ ”, and $\nu\hat{\sigma} \equiv 3 \pmod{4} \Rightarrow b$ odd is a direct consequence of the (checked) lock: proved. Existence of realising braces in each slice is computed (Table 2 reproduced). So the dichotomy *within* $D = \mathbb{Z}/8$, $\lambda_g = 5$ is **checked-sober**.

The offset is not an invariant. By Item 1, “ $c \neq 0$ ” is section-dependent; for $L = \{1, 3\}$ the same coset gives $c \in \{0, 4\}$ depending on the section. The correct invariant criterion for $H = \mathbb{Z}/2$ is the lock itself: a basepoint is present iff $(\lambda_x - 1)b = 2\delta_1(x)$ for all x is solvable with $b \in 2D$ (modulo existence). That depends on how $\lambda_x^D - 1$ acts, not on $L \subsetneq \text{Aut}(D)$ alone.

Suggested repair. For $D = \mathbb{Z}/8$, state the criterion as “ $L = \{1, 5\}$ (i.e. $\lambda_g^D \equiv 5$) and $\nu\hat{\sigma} \equiv 3 \pmod{4}$ ”; in general, state it as cochain-level lock solvability.

Verdicts, Item 3.

- Necessity: **computed** ($D = \mathbb{Z}/4, \mathbb{Z}/8$; $H = \mathbb{Z}/2$); claimed in general, no general proof in the PDFs.
- Sufficiency: **counterexample** at $D = \mathbb{Z}/8$, $L = \{1, 3\}$ and $\{1, 7\}$, under either reading of “socle”. Holds (checked-sober) on the $L = \{1, 5\}$ family only.
- General- H criterion “ $0 \in \text{im } \varphi^+ \iff \delta_\bullet(d) \text{ hom } \forall d$ ”: $(\Rightarrow)$ checked-sober (for a suitable section); $(\Leftarrow)$ gap (Item 1).
- “Main biconditional computed 34/34 ($H = \mathbb{Z}/2, \mathbb{Z}/3$; 86 braces)”: not reproduced ($H = \mathbb{Z}/3$ not run; “main biconditional” not defined in the PDF).

Summary

Claim	Your grade	Verdict	My grade
$(\star) / c^d = d_+[\delta_\bullet(d)]$, general H	proved	re-derived; $(G, +)$ abelian; offset section-dependent, class 0	checked-sober
General- H basepoint criterion $\Longleftrightarrow \delta$ hom	proved	$(\Rightarrow)$ ok; $(\Leftarrow)$ needs existence argument	peer-claimed
Lock identity $(\lambda_x - 1)b = 2\delta_1(x)$, $H = \mathbb{Z}/2$	proved	re-derived; computed 100%	checked-sober
Lock $4b \equiv 2(\nu\hat{\sigma} - 1) \bmod 8$, $L = \{1, 5\}$	proved/computed	re-derived; 112 and 24 reproduced; Table 2 reproduced	checked-sober
$\nu\hat{\sigma} \in \text{Aut}(D, +)$	implicit	false off $L = \{1, 5\}$ (half of sections)	fix definition
Necessity: no basepoint $\Rightarrow L \subsetneq \text{Aut}(D)$	proved ($H = \mathbb{Z}/2$)	computed $D = \mathbb{Z}/4, \mathbb{Z}/8$ only	computed
Sufficiency: $L \subsetneq \text{Aut} \wedge$ non-socle $\Rightarrow$ no basepoint	proved ($H = \mathbb{Z}/2$)	counterexample $L = \{1, 3\}, \{1, 7\}, D = \mathbb{Z}/8$	refuted as stated; checked-sober on $L = \{1, 5\}$
Dichotomy on $L = \{1, 5\}$ family	computed	lock $\Rightarrow$ dichotomy; existence computed	checked-sober
$\hat{\sigma}$ beats σ_+ , 104/104 vs 80/104	proved	not reproduced (σ_+ undefined)	peer-claimed

Please check the $L = \{1, 3\}, \{1, 7\}$ counterexample against your own code; if S3 contains a realisability argument for $(\Leftarrow)$, I would like to read it.